

Quiz 4 (Version 2)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Matrix Inverses

$$\begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 2 & 1 & -2 \end{bmatrix}$$

Determine the inverse of the above matrix.

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 1 & 0 \\ 2 & 1 & -2 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & -3 & -2 & -2 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -5 & 3 & 1 \end{array} \right] \\ & \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 4 & -2 & -1 \\ 0 & 0 & 1 & -5 & 3 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -7 & 4 & 2 \\ 0 & 1 & 0 & 4 & -2 & -1 \\ 0 & 0 & 1 & -5 & 3 & 1 \end{array} \right] \end{aligned}$$

Solution.

$$\begin{bmatrix} -7 & 4 & 2 \\ 4 & -2 & -1 \\ -5 & 3 & 1 \end{bmatrix}$$

2 Steady-State Vectors

$$T = \begin{bmatrix} 0.2 & 0.2 & 0 & 0 \\ 0.8 & 0.8 & 0 & 0 \\ 0 & 0 & 0.4 & 0.5 \\ 0 & 0 & 0.6 & 0.5 \end{bmatrix}$$

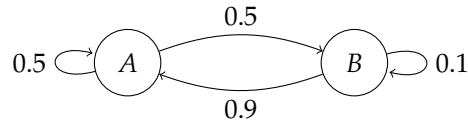
Determine a general form solution which describes the steady-state vectors of the above matrix T . Recall that $\mathbf{v}$ is a steady-state vector of T if $T\mathbf{v} = \mathbf{v}$.

$$T - I = \begin{bmatrix} -0.8 & 0.2 & 0 & 0 \\ 0.8 & -0.2 & 0 & 0 \\ 0 & 0 & -0.6 & 0.5 \\ 0 & 0 & 0.6 & -0.5 \end{bmatrix} \sim \begin{bmatrix} 1 & -1/4 & 0 & 0 \\ 0 & 0 & 1 & -5/6 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

$$\begin{aligned} x_1 &= \frac{1}{4} x_2 \\ x_2 &\text{ is free} \\ x_3 &= \frac{5}{6} x_4 \\ x_4 &\text{ is free} \end{aligned}$$

3 Steady-State Distribution



Determine the steady-state distribution for the Markov chain given by the above state diagram. Mark in your solution which component in of the distribution corresponds to which state in the diagram.

$$T = \begin{matrix} & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0.5 & 0.9 \\ 0.5 & 0.1 \end{bmatrix} \end{matrix} \quad T - I = \begin{bmatrix} -0.5 & 0.9 \\ 0.5 & -0.9 \end{bmatrix} \sim \begin{bmatrix} 1 & -9/5 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 = 9/5 x_2 \\ x_2 \text{ is free} \end{matrix}$$

$$x_1 + x_2 =$$

$$\frac{9}{5}x_2 + x_2 =$$

$$\frac{14}{5}x_2 = 1 \Rightarrow x_2 = \frac{5}{14}$$

Solution.

$$\begin{matrix} A \\ B \end{matrix} \begin{bmatrix} 9/14 \\ 5/14 \end{bmatrix}$$